



OOPS → python
→ Go lang

System Design

↓
(LLD) ↓
(MLD)

Machine Code (0100101 ~)



Assembly Language (MOV R1, R2)
(CMP X, Y)
↻



Functional / procedural programming

↳ Recipe

```
f1(int a, int b) {  
    = f2(x, y);  
}
```

```
f2(x, y) {  
}
```

Modules

```
main() {  
    = f1();  
}
```

```
f1() {  
    = f2();  
}  
f2() {  
    =  
}
```

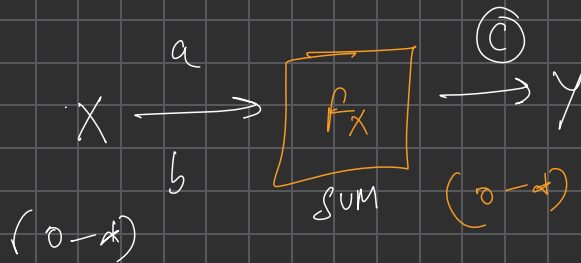
```
int main() {  
    int a;  
    int b;  
    cin >> a >> b;  
    → switch(→)  
}
```

```
int add (int a, int b) {
    return a + b;
}
```

if()
switch()
for()
while()
break, continue
function()

MC
↓
AL
↓
FL

MOV R1, R2
COM X, Y
→
→
→
→



```
Calculate (int x, int y) {
    return add, sub,
           div, mul
}
```

Object Oriented Programming

↳ "problem at scale"

Concurrent

C / Python / GoLang

OOPS

Student

↳ name
↳ age
↳ rNo
↳ College

```
String name = "Aditya";  
int age = 21;  
int rNo = 101;  
String College = "IITG";
```

```
String name = "Rohit";  
int age = 29;  
int rNo = 102;  
String College = "IITG";
```

PrintStudent()

```
cout << name << endl  
<< rNo << endl;
```

P1: Code blocks (Cannot represent real life Object)

P2: No Ownership

P3: All variables needs to be passed to a function. (printing a student)

OOPS

Why do we write Code?

↳ To solve a problem

Real world
problem

⇒ Code

Coding world.

name
age
rno
(object) college

⇒

name	age
rno	college

(Object)

```
Class Node {  
    int data;  
    Node * next;  
};
```

data	next
------	------

 →

name
age
rno
college
Student

name	age
rno	college

Abstract

Concrete

Amman
Abdulla
=

```

Class Student {
    private:
    String name;
    int age;
    int rno;
    String College;
}

```

Blueprint of an object

```

Student *s1 =
    new Student();

```

s1 → name = "Aditya"
 s1 → age = 20;
 s1 → rno = 101;
 s1 → College = "IITG";

```

Student1 *s1 = new Student();
Student *s2 = new Student();

```

s1 → name = "Aditya";

s2 → name = "Kohit";

s1 → name = "Aman"

```

PrintStudent (Student *s1) {
    cout << s1 → name << s1 → age
        << s1 → rno << s1 → College;
}

```

OOPS

```

class Student {
    String name;
    int age;
    // Constructor
    Student(String n, int a) {
        name = n;
        age = a;
    }
}

Student *s1 = new Student("Aditya", 29);
cout << s1->name;
s1->age;

Student *s2 = new Student();

```

Parameterized

```

class Student {
    int age;
    String name;
    Student() {
    }
}

Student *s1 = new Student();
s1->name = " ";
s1->age = 29;

```

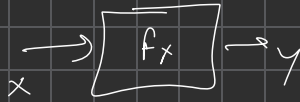
Default


 Characteristics (name, age, sex, roll-...)


Behaviours (markAttendance(), giveExam(), ...)

(Student)

○ → characters (name, age, rNo - - -)
 ✕ → Behaviour (markAttendance)
 Student



```

Class Student {
    String name;
    int age;
    int rNo;
    String College;
    // Constructors

    void markAttendance() {
        // body
    }

    void print() {
        cout << name << endl;
        << rNo << endl;
    }
}
  
```

Characters (points to attributes: name, age, rNo, College)
 Behaviour (points to methods: markAttendance, print)
 print() (highlighted in a box)